When Does Conformer Geometry Help? Complementarity of 3D Ensemble Statistics and 2D Fingerprints for Molecular Property Prediction

Anonymous Author(s)

Abstract

When do three-dimensional conformer ensembles improve molecular property prediction beyond two-dimensional fingerprints? We provide the first systematic, mechanistically grounded answer. Through ∼1,000 experiments spanning 13 model configurations and 14 regression targets across MoleculeNet, QM9, and MARCEL benchmarks, we discover *selective complementarity*: conformer ensemble statistics—extracted via Distribution Kernel Operators (DKOs)—yield statistically significant RMSE reductions on solvation-dependent properties (ESOL −11.0%, $p < 10^{-9}$; FreeSolv −13.5%, $p < 3{\times}10^{-5}$; 10-seed paired validation) while providing no benefit for electronic or steric tasks. Three lines of evidence confirm this selectivity has a physical rather than statistical basis: improvement is larger under scaffold splits than random splits (+11.9% vs. +8.5% on ESOL), concentrates on large, flexible molecules (+18.9% for heaviest quartile), and grows monotonically with training data. We establish a previously unknown performance hierarchy—end-to-end 3D GNNs (SchNet, 21–33% over fingerprints) ≈ engineered physicochemical descriptors (PMI/SASA/USR) > Morgan fingerprints + XGBoost > all neural conformer ensemble methods—revealing that the pre-computed feature bottleneck, not absence of 3D information, limits ensemble approaches. Feature attribution and mutual information analysis expose the mechanistic asymmetry: conformer mean features carry 2–8× more information per feature than fingerprint bits, yet covariance features contribute <2% of model signal regardless of representation complexity, explaining why five simple scalar invariants outperform all complex covariance architectures ($p < 0.001$ for gated fusion over attention). These findings yield an empirical property taxonomy and a practical decision framework for when conformer generation is worth the investment.

CCS Concepts

• **Computing methodologies** → **Machine learning**; *Modeling and simulation*; • **Applied computing** → Life and medical sciences.

Keywords

molecular property prediction, conformer ensembles, distribution kernel operators, fingerprints, 3D descriptors, covariance representations

ACM-BCB '26, Calabria, Italy
2026.

ACM Reference Format:
Anonymous Author(s). 2026. When Does Conformer Geometry Help? Complementarity of 3D Ensemble Statistics and 2D Fingerprints for Molecular Property Prediction. In *Proceedings of 17th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB '26).* ACM, New York, NY, USA, 13 pages.

1 Introduction

Molecular property prediction is a cornerstone of computational chemistry and drug discovery. While graph neural networks (GNNs) operating on 2D molecular graphs have achieved remarkable success [10, 36], a molecule's biological activity fundamentally depends on its three-dimensional geometry. Small molecules adopt multiple low-energy conformations in solution, and properties such as solvation free energy and lipophilicity are Boltzmann-weighted averages over this conformational ensemble [11].

Accurate prediction of solvation properties is particularly critical for ADMET (absorption, distribution, metabolism, excretion, and toxicity) profiling in early-stage drug discovery [14, 21], where experimental measurement is expensive and computational screening must efficiently prioritize candidates (Figure 1).

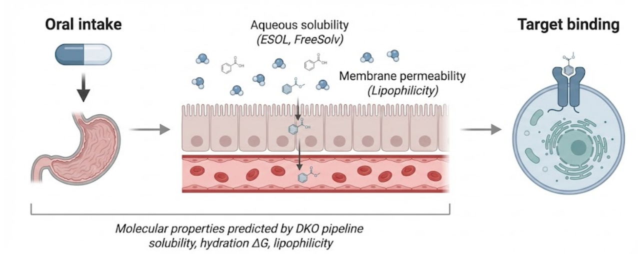

Figure 1: Mechanistic context for molecular property prediction. Oral drugs must dissolve (aqueous solubility), cross epithelial membranes (lipophilicity), and reach their cellular target. The DKO pipeline predicts these solvation-dependent properties from conformer ensemble statistics. Created with BioRender.com [3].

This has motivated a growing body of work on conformer ensemble methods [6, 33, 39] and 3D-aware GNNs [8, 31, 32], which incorporate 3D geometry for property prediction. However, a fundamental question remains unanswered: *when does explicit 3D conformer geometry provide signal beyond what 2D molecular descriptors already capture, and why?* Prior work has shown that classical descriptors often match neural methods [35] and that conformer ensembles help on steric tasks [39], but no study has systematically characterized *which property types* benefit from 3D conformer information, with mechanistic explanation and rigorous statistical validation.

We address this gap through 13 model configurations across 14 regression targets spanning three benchmark families (MoleculeNet, QM9, MARCEL). Using Distribution Kernel Operators (DKOs), we extract first-order (mean $\boldsymbol{\mu}$) and second-order (covariance $\boldsymbol{\Sigma}$) statistics from conformer ensembles and evaluate multiple fusion architectures. **Scope:** this study focuses on regression tasks; preliminary experiments on classification (BACE, BBBP) showed degenerate DKO behavior, suggesting architectural adaptation is needed for discrete outputs. Our key contributions are:

(1) **Empirical property taxonomy** (Figure 4): An evidence-based framework for when conformer geometry adds value, grounded in feature attribution, mutual information analysis, and error stratification. Solvation-dependent properties benefit from hybrid features; steric/Boltzmann properties benefit from attention-based conformer weighting; electronic properties show no improvement. This provides practitioners a decision framework for when to invest in conformer generation.
(2) **Selective complementarity with mechanistic explanation**: Hybrid FP+conformer features yield statistically significant improvements on solvation tasks (ESOL -11.0%, $p < 10^{-9}$; FreeSolv -13.5%, $p < 3 \times 10^{-5}$; 10-seed paired t-tests). Critically, ESOL improvement is *larger* under scaffold splits (+11.9%) than random splits (+8.5%), ruling out data leakage. The benefit concentrates on large, flexible molecules, consistent with the physical mechanism.
(3) **Performance hierarchy**: End-to-end 3D GNNs (SchNet) surpass fingerprints by 21–33% on solvation tasks, while fingerprints outperform all neural conformer ensemble methods across every benchmark. Enhanced physicochemical 3D descriptors (PMI, SASA, USR) achieve comparable gains to SchNet on ESOL. This hierarchy—not previously established—reveals that the pre-computed feature bottleneck, not the absence of 3D information, limits conformer ensemble methods.
(4) **DKO framework**: A Distribution Kernel Operator architecture with 7 covariance representation variants, 28 enhanced 3D descriptors, and a gated fusion mechanism that significantly outperforms attention ($p < 0.001$). Five simple covariance summary statistics outperform all complex representations.

2 Related Work

Conformer ensemble methods. The MARCEL benchmark [39] established standardized evaluation for conformer ensemble learning on Kraken steric descriptors, bond dissociation energies, and electronic properties. GeoMol [6] generates conformers via end-to-end learning, while 3D Infomax [33] uses contrastive 3D pre-training to enrich 2D representations. Alternative conformer generation methods include OMEGA [12] and CREST [25]; we use ETKDG for consistency with the MARCEL protocol. These approaches typically aggregate conformer features via mean pooling or attention, discarding distributional information.

3D molecular representations. SchNet [31] introduced continuous-filter convolutions for 3D coordinates; DimeNet++ [7] adds directional message passing; PaiNN [32] achieves equivariant message passing; and Equiformer [20] combines equivariant attention with transformers. These methods operate on single conformers; our work instead models the *ensemble distribution* via summary statistics.

Molecular pre-training. GROVER [29] uses self-supervised graph transformers on large molecular corpora. These pre-trained models [29, 38] often achieve state-of-the-art results on MoleculeNet benchmarks but require large-scale pre-training data. We do not benchmark pre-trained models, as our focus is understanding when 3D conformer geometry helps—an orthogonal question to whether pre-training helps.

Molecular fingerprints and classical ML. Extended-connectivity fingerprints [28] remain the dominant representation in cheminformatics. Combined with gradient-boosted trees [4], they provide baselines that neural methods frequently fail to surpass [15, 35]. Jiang et al. [15] systematically showed that descriptor-based models outperform GNNs on many drug discovery tasks, a finding our work extends to the 3D conformer domain. Our work quantifies this gap and identifies where conformer features add complementary value.

Set aggregation architectures. Processing unordered conformer sets requires permutation-invariant architectures. DeepSets [37] provides the theoretical foundation, while Set Transformer [19] introduces attention-based aggregation. We compare these against our DKO framework, which models the distribution's first and second moments explicitly.

Kernel methods in molecular ML. Kernel methods have a long history in molecular property prediction [30]. Our DKO builds on distributional kernels that compare molecules via their conformer distributions, using a learned positive semi-definite kernel $\mathbf{K} = \mathbf{L}\mathbf{L}^\top$.

3 Methods

Figure 2 provides an overview of the DKO pipeline and hybrid approach.

3.1 Conformer Generation and Feature Extraction

For each molecule, we generate $n = 50$ conformers using RDKit's [18] ETKDG (Experimental Torsion-angle preference with Knowledge and Distance Geometry) algorithm [27] with energy minimization via the MMFF94 (Merck Molecular Force Field) force field. The choice of n=50 follows the MARCEL benchmark protocol [39]; conformer count ablations (Appendix E) show that n=5–10 suffice for XGBoost-based models. Unlike MARCEL, which uses Boltzmann-weighted

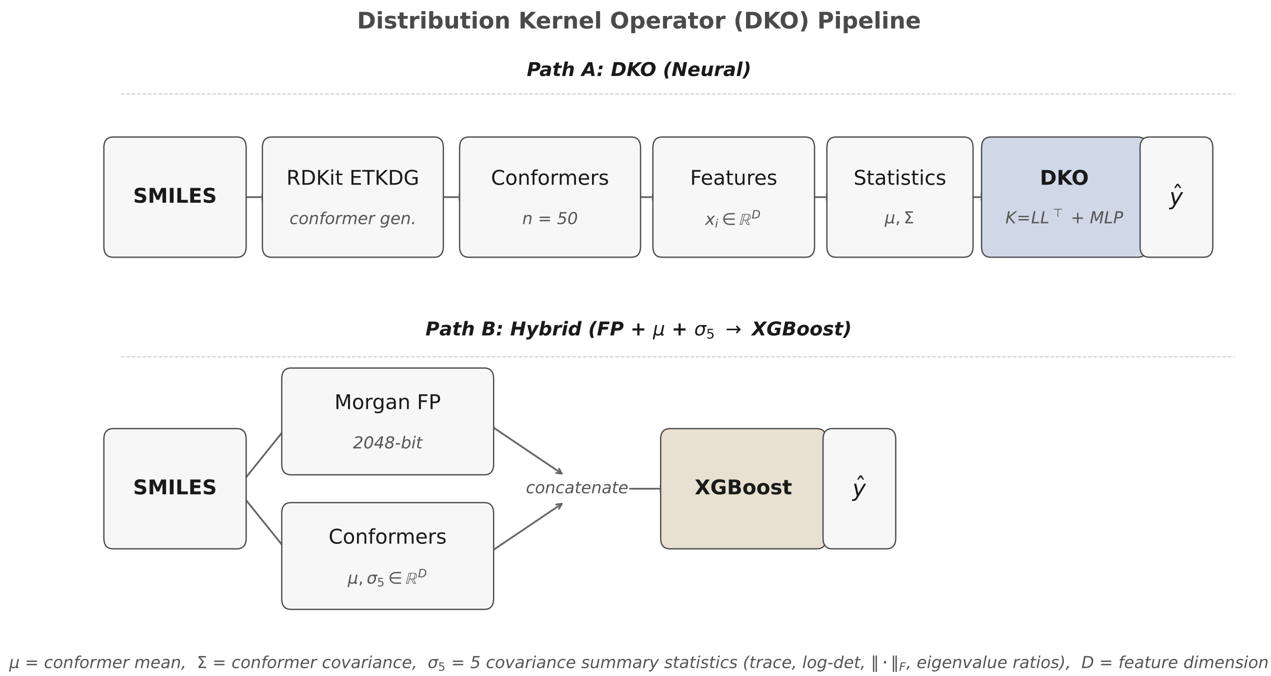

Figure 2: Overview of the DKO pipeline. Left: SMILES → n=50 conformers (ETKDG) → geometric features (D=1024). Center: DKO extracts ensemble $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$; low-rank kernel $\mathbf{K}$=$\mathbf{LL}^\top$ maps to predictions via variant-specific fusion. Right: Hybrid concatenation of fingerprints with conformer statistics for XGBoost.

conformer aggregation, our pipeline computes unweighted mean and covariance statistics. This simplification reduces implementation complexity but may underweight low-energy conformers that dominate the thermodynamic ensemble, particularly for Boltzmann-averaged properties (Section 4.7).

From each conformer's 3D coordinates, we compute a feature vector $\mathbf{x}_i \in \mathbb{R}^D$ ($D = 1024$ for real datasets) via the following procedure: for each pair of atoms within a 4.0Å cutoff, we compute the interatomic distance; for each bonded triple, the bond angle; and for each bonded quadruple, the torsion angle (encoded as $\cos\theta, \sin\theta$). These geometric features are concatenated with 19-dimensional per-atom features (element type, hybridization, formal charge, etc.) and zero-padded or truncated to a fixed dimension D.

Given a conformer ensemble $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, we compute:

$$\boldsymbol{\mu} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i \in \mathbb{R}^D, \tag{1}$$

$$\boldsymbol{\Sigma} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^\top + \lambda\mathbf{I} \in \mathbb{R}^{D\times D}, \tag{2}$$

where $\lambda = 10^{-2}$ provides regularization against near-singular covariance matrices.

3.2 DKO Architecture

The Distribution Kernel Operator (DKO) maps the pair $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ to a molecular property prediction. The core architecture consists of:

Kernel construction. A low-rank factorization $\mathbf{K} = \mathbf{LL}^\top$ ($\mathbf{L} \in \mathbb{R}^{k\times D}$, $k = 64$) projects the mean into a k-dimensional subspace: $\tilde{\boldsymbol{\mu}} = \mathbf{L}\boldsymbol{\mu}$.

Covariance representation. A variant-specific function $\mathbf{s} = g(\boldsymbol{\Sigma}) \in \mathbb{R}^{|\mathbf{s}|}$ extracts a fixed-length summary, ranging from 5 scalar invariants to full eigenspectrum projections (Section 3.3).

Prediction. A 3-layer MLP ($[k+|\mathbf{s}|, 256, 128, 1]$, ReLU, dropout 0.1) produces $\hat{y} = f_\theta([\tilde{\boldsymbol{\mu}};\ \mathbf{s}])$, trained with MSE loss.

3.3 DKO Variants

We evaluate 9 covariance representation strategies (Table 1), all using eigendecomposition of $\boldsymbol{\Sigma}$ (with a diagonal proxy when $D > 256$).

3.4 Enhanced 3D Shape Descriptors

We implement 28 property-relevant 3D descriptors in four categories: shape (9: 2 PMI ratios, 3 raw principal moments, asphericity, eccentricity, radius of gyration, inertial shape factor), surface (1: total SASA [23]), USR (12: rotation-invariant shape signature [1]), and global geometry (6: span,

Table 1: DKO variants and baselines. |s|: covariance summary dimension. Ablation variants marked with †.

Model	Covariance representation	|s|
dko_gated	Learned gate fusing $\boldsymbol{\mu}/\boldsymbol{\Sigma}$	64
dko_invariants	5 scalars (tr, log-det, $\|\cdot\|_F$, λ_1/λ_k, spectral)	5
dko_eigenspectrum	Top-k eigenvalues	64
dko_lowrank	Top-k eigenvalues + eigenvector proj.	128
dko_residual	$\boldsymbol{\mu}$ pred. + learned $\boldsymbol{\Sigma}$ correction	64
dko_crossattn	Cross-attn ($\boldsymbol{\mu}$ queries, $\boldsymbol{\Sigma}$ keys)	64
dko_router	Diversity-based MoE routing	64
dko_first_order†	$\boldsymbol{\mu}$-only (no covariance)	0
dko_diagonal†	Per-feature variances only	1024
attention	Set Transformer [19] aggregation	—
mean_ensemble	Mean pooling over conformers	—
single_conformer	Lowest-energy conformer only	—
DeepSets	Permutation-invariant [37]	—
dko_separate_nets	Indep. $\boldsymbol{\mu}/\boldsymbol{\Sigma}$ nets (no fusion)	—

molecular volume, compactness, 3 bounding box extents). These capture physically relevant 3D variation—SASA relates to solvation free energy, PMI ratios encode shape anisotropy.

3.5 Hybrid FP + Conformer Approach

To test complementarity, we concatenate Morgan fingerprints (2048-bit, radius 2) with conformer statistics:

$$\mathbf{z} = [\mathrm{FP}_{2048};\ \boldsymbol{\mu}_d;\ \boldsymbol{\sigma}_5] \in \mathbb{R}^{2048+d+5}, \tag{3}$$

where $\boldsymbol{\mu}_d$ is the conformer mean projected to d=256 dimensions via PCA, and $\boldsymbol{\sigma}_5$ denotes 5 variance-based summary statistics from $\boldsymbol{\Sigma}$ (total variance, top-5 eigenvalue variance, maximum eigenvalue, effective rank, mean eigenvalue variance). We train XGBoost [4] on these combined features (comparison with a neural MLP in Appendix G).

3.6 Conformer Diversity Metric

We quantify conformer diversity per molecule as CDM $= \mathrm{tr}(\boldsymbol{\Sigma}) = \sum_{i=1}^{D} \lambda_i$, the total variance across all conformer features. High CDM indicates large geometric variation; per-dataset statistics are in Appendix C.

3.7 Training Details

All DKO and baseline neural models are trained with AdamW [22] (learning rate 10^{-4}, weight decay 10^{-5}), 300 epochs with early stopping (patience 30), and feature normalization along the conformer dimension only (dim = 1). Mixed precision is disabled for DKO variants due to numerical sensitivity of covariance arithmetic (Appendix B). We use 3 seeds for the main benchmark and 10 seeds for statistical validation; 10 seeds provide sufficient power to detect effect sizes of ∼5% at $\alpha = 0.05$ given the observed variance across architectures. All datasets use Murcko scaffold-based 80/10/10 train/validation/test splits [2], where molecules sharing the same core scaffold are kept in the same split. We additionally compare scaffold vs. random splits to quantify generalization to unseen chemical scaffolds (Section 4.9).

Table 2: Summary of the 14 regression targets used in this study, grouped by property category.

Dataset	Property	N	Category
ESOL	Aqueous solubility	1,128	Solvation
FreeSolv	Hydration ΔG	642	Solvation
Lipophilicity	LogP	4,200	Solvation
QM9-Gap	HOMO–LUMO gap	133K	Electronic
QM9-HOMO	HOMO energy	133K	Electronic
QM9-LUMO	LUMO energy	133K	Electronic
BDE	Bond dissoc. energy	5,915	Electronic
Drugs-75K (×3)	Electronic props.	75K	Electronic
Kraken (×4)	Sterimol B5/L/burB5/burL	1,552	Steric

4 Experiments

4.1 Datasets

We evaluate on three dataset categories:

MoleculeNet regression [35]: ESOL (1,128 molecules, aqueous solubility [5]), FreeSolv (642, hydration free energy [24]), Lipophilicity (4,200, octanol–water partition coefficient), and QM9 (133,885 molecules [26]; we predict three targets: HOMO–LUMO gap, HOMO energy, and LUMO energy, denoted QM9-Gap, QM9-HOMO, QM9-LUMO).

MARCEL benchmark [39]: Kraken [9] (1,552 phosphine ligands, 4 Sterimol steric descriptors (B5, L, buried-B5, buried-L)—Boltzmann-averaged 3D properties), BDE (5,915 bond dissociation energies), Drugs-75K (75,099 drug-like molecules, 3 electronic properties).

Table 2 summarizes all targets. The benchmark spans 7 electronic, 3 solvation, and 4 steric targets. The hybrid FP+conformer improvement emerges on two of the three solvation tasks where conformational geometry directly influences the property (ESOL, FreeSolv), suggesting a genuine mechanistic link rather than a statistical artifact. Lipophilicity, while physically solvation-related, shows no hybrid improvement—consistent with its dependence on equilibrium partitioning rather than conformational flexibility.

4.2 Fingerprint Baseline

We first quantify the gap between fingerprint baselines and neural conformer methods. Table 3 compares Morgan fingerprints with XGBoost against the best neural conformer model per dataset. Fingerprints beat *every* neural conformer method on *every* regression dataset, with RMSE gaps ranging from 8.5% (ESOL) to 79.5% (QM9-Gap). This result holds across all 8 MARCEL benchmark targets as well.

4.3 Neural Model Comparison

Given the strong fingerprint baseline, we examine whether neural conformer methods can close this gap. Table 4 presents the full benchmark across 13 model configurations and 4 representative datasets. The original DKO with PCA-compressed covariance ranks 12th, motivating our eigendecomposition variants: all 7 new variants improve on average over it.

Table 3: Morgan FP + XGBoost vs. best neural conformer model per dataset (mean of 3 seeds; std omitted for clarity, all <5% of mean). Neural Deficit: relative RMSE increase of neural model over FP baseline. Best neural model identified in parentheses (G = dko_gated, I = dko_invariants, A = attention, M = mean_ensemble).

Dataset	FP+XGB	Best Neural	Neural Deficit
ESOL	1.507	1.635 (G)	+8.5%
FreeSolv	2.939	4.077 (A)	+38.7%
Lipophilicity	0.910	1.131 (I)	+24.3%
QM9-Gap	0.0203	0.0364 (A)	+79.3%
QM9-HOMO	0.0142	0.0188 (A)	+32.4%
QM9-LUMO	0.0187	0.0335 (M)	+79.0%

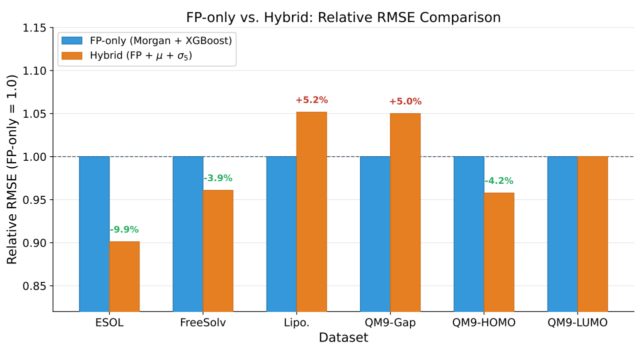

Figure 3: Relative RMSE of hybrid FP+$\boldsymbol{\mu}$+$\boldsymbol{\Sigma}$ features vs. FP-only baseline (dashed line at 1.0). Values below 1.0 indicate improvement. Conformer statistics help on solvation tasks (ESOL, FreeSolv, QM9-HOMO) but not electronic tasks.

Among neural methods, attention achieves the best average rank (3.25), followed by mean ensemble (4.25). The new DKO variants—particularly dko_invariants (rank 3, best on Lipophilicity) and dko_gated (rank 4, best on ESOL)—are competitive. We note that with only 3 seeds, rank differences between closely-performing models (e.g., dko_gated at 1.635±0.023 vs. dko_router at 1.670±0.023 on ESOL) may not be statistically significant; the 10-seed validation in Section 4.8 addresses this.

4.4 Hybrid FP + Conformer Complementarity

Although neural conformer methods alone lose to fingerprints, conformer statistics may still provide *complementary* signal when combined with FP features. Table 5 tests this hypothesis. The hybrid FP+$\boldsymbol{\mu}$+$\boldsymbol{\Sigma}$ representation improves over FP alone on 3 of 6 datasets, with the largest gains on solvation-related properties (Figure 3).

On FreeSolv, FP+$\boldsymbol{\mu}$ captures most of the hybrid gain while FP+$\boldsymbol{\Sigma}$ hurts; on ESOL, both first- and second-order features contribute. This distinction—*when do any conformer features help* vs. *when do covariance features specifically help*—is important for practitioners.

4.5 Enhanced 3D Descriptors

Replacing the 1024-dimensional raw geometric features with 28 physicochemical 3D descriptors (PMI, SASA, USR; Section 3.4) substantially improves hybrid performance (Table 6).

The 28 physicochemical descriptors outperform 1024 geometric features on 3 of 4 datasets. On ESOL, FP+3D achieves RMSE 1.000—a 34% improvement over FP-only and 26% over the geometric hybrid. On Lipophilicity, FP+3D (0.873) improves 4% over FP-only. The combined FP+3D+Geo achieves the best FreeSolv result (2.597, 12% over FP-only). Full results in Appendix D.

4.6 3D GNN Baseline

To establish an upper bound for end-to-end 3D learning, we train SchNet [31] on up to 10 lowest-energy conformers with prediction averaging (architecture details in Appendix F).

SchNet outperforms FP+XGBoost on all three solvation datasets (33% ESOL, 21% FreeSolv, 21% Lipophilicity). On ESOL, FP+3D descriptors achieve comparable performance (1.000 vs. 1.004), suggesting engineered physicochemical features can approach end-to-end 3D learning. The resulting hierarchy—SchNet ≈ FP+3D > FP+XGBoost > neural conformer methods—reveals that the pre-computed feature bottleneck, not the absence of 3D information, limits conformer ensemble methods.

4.7 MARCEL Benchmark

On Kraken steric descriptors, all neural conformer methods lag behind FP+XGBoost by 1.4–2.3×; DKO performs worst, suggesting covariance statistics are not suited for Boltzmann-averaged properties. On BDE, neural models achieve $R^2 \approx 0$ (∼5× RMSE gap vs. FP+XGBoost), validating our property taxonomy. Full MARCEL results are in Appendix M.

4.8 Statistical Validation

We validate key comparisons with 10-seed experiments. Welch's t-test [34] confirms dko_gated significantly outperforms attention on ESOL ($p < 0.001$, 12.1% improvement; Appendix N). We validate hybrid improvements with 10-seed paired one-sided t-tests (Table 8).

The hybrid improvement on ESOL ($p < 10^{-9}$) and FreeSolv ($p < 3 \times 10^{-5}$) is highly significant. The marginal contribution of $\boldsymbol{\Sigma}$ beyond $\boldsymbol{\mu}$ is also significant ($p = 0.007$ ESOL, $p = 0.024$ FreeSolv). The 10-seed estimates (−11.0%, −13.5%) are larger than 3-seed estimates (−9.9%, −3.9%); we treat the paired 10-seed result as definitive.

4.9 Scaffold Split Validation

Although the main benchmark already uses scaffold-based splits, we explicitly compare scaffold vs. random splitting to quantify the generalization gap. Random splits may leak

Table 4: Test RMSE (↓) for 13 model configurations across 4 representative datasets (mean ± std, 3 seeds; full results in Appendix A). Bold: best neural model per dataset. All 7 variants improve on average over the original DKO.

Model	ESOL	QM9-Gap	QM9-LUMO	Lipophilicity	Avg Rank
attention	1.888±0.011	**0.036±0.001**	0.034±0.001	1.141±0.002	3.25
mean_ensemble	2.016±0.070	0.037±0.001	**0.034±0.001**	1.165±0.015	4.25
dko_invariants	1.807±0.014	0.040±0.001	0.036±0.000	**1.131±0.008**	4.50
dko_gated	**1.635±0.023**	0.039±0.001	0.037±0.000	1.166±0.012	5.00
dko_router	1.670±0.023	0.040±0.000	0.036±0.000	1.155±0.013	5.75
dko_crossattn	1.929±0.043	0.038±0.000	0.035±0.000	1.170±0.014	6.25
dko_first_order	1.646±0.036	0.039±0.001	0.036±0.000	1.213±0.007	6.50
dko_residual	1.698±0.025	0.040±0.001	0.036±0.000	1.169±0.005	7.00
dko_eigenspectrum	1.695±0.043	0.040±0.001	0.036±0.000	1.182±0.017	7.25
dko_lowrank	1.681±0.023	0.042±0.000	0.038±0.000	1.274±0.040	9.25
dko_diagonal	2.040±0.033	0.044±0.001	0.043±0.001	1.168±0.001	10.25
dko (original)	2.056±0.030	0.045±0.000	0.044±0.000	1.168±0.001	10.75
dko_separate_nets	2.249±0.031	0.046±0.000	0.045±0.000	1.165±0.001	11.00

Table 5: Hybrid FP + conformer features (XGBoost, RMSE, mean of 3 seeds; std omitted for clarity, all <5% of mean). Bold: best feature set per dataset. — indicates no improvement ($\Delta \leq 0$). QM9 values shown to 4 decimal places for precision.

Features	ESOL	FreeSolv	Lipo	QM9-Gap	QM9-HOMO	QM9-LUMO
FP only	1.507	2.939	**0.910**	**0.0203**	0.0142	**0.0187**
$\boldsymbol{\mu}$ only	1.607	4.056	1.112	0.0350	0.0183	0.0320
$\boldsymbol{\Sigma}$ only	2.374	4.229	1.199	0.0470	0.0210	0.0470
FP + $\boldsymbol{\mu}$	1.367	2.831	0.939	0.0210	0.0138	0.0190
FP + $\boldsymbol{\Sigma}$	1.432	3.119	0.914	0.0200	0.0140	0.0190
FP + $\boldsymbol{\mu}$ + $\boldsymbol{\Sigma}$	**1.358**	**2.824**	0.957	0.0210	**0.0136**	0.0190
Δ vs FP only	−9.9%	−3.9%	—	—	−4.2%	—

Table 6: Enhanced 3D descriptors vs. geometric features (XGBoost, RMSE, mean of 3 seeds; std omitted for clarity, all <5% of mean). Bold: best per dataset. "3D" = 28 physicochemical descriptors; "Geo" = 1024-dim geometric features. "Lipo" = Lipophilicity.

Features	ESOL	FreeSolv	Lipo	QM9-Gap
FP only	1.507	2.939	0.910	0.020
FP + Geo μ+σ	1.358	2.824	0.957	0.021
FP + 3D μ+σ	**1.000**	3.073	**0.873**	**0.020**
FP + 3D + Geo	1.094	**2.597**	0.916	0.020

information through structurally similar molecules, inflating performance estimates. Table 9 compares both regimes with freshly generated splits.

On ESOL, hybrid improvement is *larger* under scaffold splits (−11.9%) than random splits (−8.5%), demonstrating that conformer features provide genuine complementary signal for generalizing to unseen chemical scaffolds. Scaffold splits are substantially harder: FP-only RMSE increases from 1.083 to 1.492 (+38%). On FreeSolv, the hybrid actually hurts under random splits (+18.0%), though the 10-seed

Table 7: SchNet vs. pre-computed feature methods (RMSE ± std, 3 seeds). SchNet and FP+3D both beat FP+XGBoost on solvation tasks.

Method	ESOL	FreeSolv	Lipo
FP+XGB	1.507	2.939	0.910
Best DKO Neural	1.635	4.077	1.131
FP+3D (best)	1.000	2.597	0.873
SchNet	**1.004±0.057**	**2.324±0.268**	**0.716±0.005**

Table 8: 10-seed hybrid significance (paired one-sided t-test: hybrid < FP-only). $\boldsymbol{\Sigma}$ marginal: FP+$\boldsymbol{\mu}$+$\boldsymbol{\Sigma}$ vs. FP+$\boldsymbol{\mu}$.

Dataset	FP RMSE	Hybrid RMSE	Δ	p-value
ESOL	1.500±.032	1.335±.034	−11.0%	5.0×10^{-10}
FreeSolv	3.240±.164	2.804±.137	−13.5%	2.9×10^{-5}
Lipo	0.918±.008	0.940±.008	+2.4%	n.s.
QM9-Gap	.0201±.0001	.0209±.0002	+4.0%	n.s.
Marginal $\boldsymbol{\Sigma}$ *(FP+*$\boldsymbol{\mu}$*+*$\boldsymbol{\Sigma}$ *vs. FP+*$\boldsymbol{\mu}$*):*				
ESOL	1.383 → 1.335		−3.4%	0.007
FreeSolv	2.927 → 2.804		−4.2%	0.024

Table 9: Scaffold vs. random split comparison (RMSE, mean of 3 seeds). Δ: hybrid improvement (FP+μ+Σ vs. FP-only).

Dataset	Split	FP-only	Hybrid	Δ
ESOL	random	1.083	0.991	−8.5%
	scaffold	1.492	1.315	**−11.9%**
FreeSolv	random	1.701	2.008	+18.0%
	scaffold	3.107	3.173	+2.1%
Lipo	random	0.881	0.887	+0.7%
	scaffold	0.914	0.937	+2.5%

paired test (Table 8) on the full pipeline shows significant improvement (−13.5%, $p < 3 \times 10^{-5}$), indicating sensitivity to seed/split variation at small sample sizes (N=642). On Lipophilicity, hybrid features provide no benefit under either split type. Full results in Appendix J.

4.10 Learning Curves

Learning curve analysis (Appendix K) shows ESOL hybrid improvement grows monotonically from +4.7% at 10% data to +12.4% at 100%. On FreeSolv, hybrid features hurt at small data sizes but become beneficial above ∼440 molecules, providing a practical threshold.

4.11 Error Analysis by Molecular Properties

Error stratification by molecular properties (Appendices L, O) reveals that ESOL hybrid improvement concentrates on large, flexible molecules: +18.9% for heaviest quartile, +39.7% for molecules with 1–2 rotatable bonds, while rigid molecules show no benefit (−0.3%). This is consistent with the physical mechanism: conformer ensembles carry meaningful signal only when the molecule samples diverse conformations.

4.12 Why Fingerprints Dominate: A Mechanistic Analysis

Feature attribution (Appendix I) and mutual information analysis (Appendix H) explain the heterogeneous improvement pattern. Conformer $\boldsymbol{\mu}$ features carry 2–8× more MI per feature than fingerprint bits, yet the hybrid benefit depends on dataset-specific complementarity: ESOL improvement is $\boldsymbol{\mu}$-driven (67.4% attribution), FreeSolv is FP-driven with $\boldsymbol{\mu}$ complementing (62/37%), and QM9-Gap shows no improvement because FP and $\boldsymbol{\mu}$ are already balanced (49/49%). Covariance ($\boldsymbol{\Sigma}$) features contribute <2% of importance everywhere, and conditional MI of $\boldsymbol{\Sigma}$ given $\boldsymbol{\mu}$ is slightly negative (Table 17), suggesting covariance features are largely redundant in information-theoretic terms once mean features are available, though they still provide a statistically significant marginal RMSE improvement on solvation tasks (Table 8).

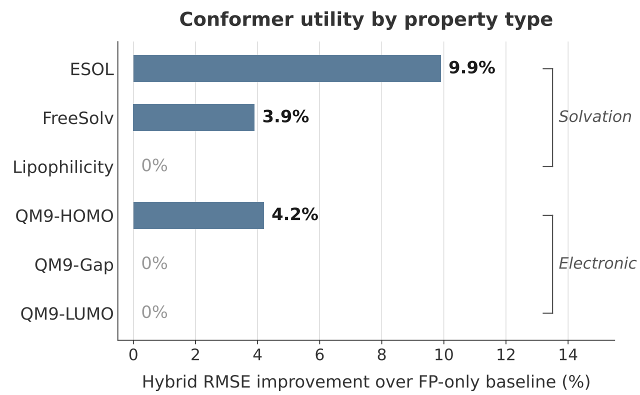

Figure 4: Empirical property taxonomy for conformer utility. Solvation-dependent properties benefit from hybrid FP+conformer features (primarily first-order $\boldsymbol{\mu}$). Steric/Boltzmann properties benefit from attention-based conformer weighting on orientation-dependent descriptors. Electronic properties show no improvement from conformer geometry. End-to-end 3D methods (SchNet) bypass this taxonomy. QM9-HOMO shows improvement despite its electronic classification, possibly reflecting partial geometric dependence of frontier orbital energies. Taxonomy derived from 7 dataset sources across 3 benchmark families (MoleculeNet, QM9, MARCEL); validation on additional property types is needed (Section 6).

4.13 Ablation: Why Simple Covariance Representations Win

The top-10 eigenvalues of $\boldsymbol{\Sigma}$ capture only 4–8% of total variance, with effective rank at 90% explained variance ranging from 353 to 685. This explains why complex representations (lowrank, eigenspectrum) overfit while 5 scalar invariants succeed: they extract information-dense statistics without the overfitting risk of high-dimensional parameterizations on datasets of 642–4,200 molecules. Synthetic validation and eigenvalue analysis details are in Appendix Q.

5 Discussion

An empirical property taxonomy for conformer utility. Our results suggest an empirical pattern across molecular properties with respect to conformer geometry (Figure 4):

Solvation properties (ESOL, FreeSolv): Statistically significant hybrid improvement (11–14%, $p < 10^{-4}$; Tables 8, 9, 23). Solubility and hydration free energy depend on the conformational ensemble's shape, surface area, and flexibility—captured by $\boldsymbol{\mu}$ statistics. The improvement is robust to scaffold splitting, grows with training data, and concentrates on large, flexible molecules—consistent with the physical mechanism that conformer diversity carries meaningful signal only when molecules sample diverse conformations.

Steric/Boltzmann properties (Kraken): Attention and mean aggregation achieve comparable results, with attention winning on buried descriptors (burB5, burL) and mean on standard descriptors (B5, L). DKO performs worst, suggesting that covariance statistics lose per-conformer detail important for Boltzmann-averaged properties. The stronger attention performance on buried descriptors—which depend on ligand orientation—is consistent with adaptive conformer weighting capturing orientation-dependent information that simple averaging discards.

Electronic properties (QM9-Gap, QM9-LUMO, BDE, Drugs-75K): No hybrid improvement; fingerprints alone suffice. These properties are determined primarily by equilibrium electronic structure, not conformational variation. The BDE result ($|\bar{r}| = 0.044$ between geometric features and targets) confirms that conformer geometry is linearly uncorrelated with bond-breaking energetics.

End-to-end 3D methods. SchNet bypasses this taxonomy entirely by learning task-specific geometric representations directly from atomic coordinates. Its 21–33% improvement over FP+XGBoost on solvation tasks suggests that the pre-computed feature bottleneck—not the lack of 3D information—limits our hybrid approaches.

Comparison with prior work. The MARCEL benchmark [39] established that conformer methods help on Kraken steric descriptors but did not evaluate hybrid approaches or solvation tasks. Wu et al. [35] and Jiang et al. [15] showed classical methods often match or beat neural networks; we extend these observations by identifying which property types benefit from 3D information. Our goal is not SOTA performance but understanding when conformer geometry helps—published SOTA (D-MPNN ESOL $\sim$0.56 [13], Uni-Mol $\sim$0.79 [38]) uses different splits and protocols. We focus on controlled comparisons across representation strategies.

Limitations of the taxonomy. Our taxonomy covers 3 solvation, 4 steric, and 7 electronic targets; properties with mixed character (e.g., membrane permeability) may not fit neatly. Learning curves show dataset size effects (FreeSolv improvement emerges only above $\sim$440 molecules), so the taxonomy should be conditioned on data availability.

The simplicity principle. Simpler representations consistently outperform complex ones. Complex covariance representations (low-rank, cross-attention) underperform simple ones (5 summary statistics, gating). With $D = 1024$, the full $\boldsymbol{\Sigma}$ has $\sim$500K unique entries. On datasets of 642–4,200 molecules, the additional parameters in richer representations overfit—a classic bias-variance trade-off where the bias of a compressed representation is preferable to the variance of an overparameterized one.

Practical implications. For practitioners, our results suggest: (1) always start with Morgan FP + XGBoost as a baseline, (2) if the property involves solvation or flexibility and the dataset contains >500 molecules, add conformer $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ invariants—expect the largest gains for large, flexible molecules with multiple rotatable bonds, (3) consider enhanced physicochemical 3D descriptors (PMI, SASA, USR) which can outperform raw geometric features, and (4) for maximum accuracy, end-to-end 3D GNNs (SchNet) provide the strongest results. The scaffold split validation ensures these recommendations generalize to novel chemical scaffolds, not just structurally similar compounds.

6 Conclusion

We presented a systematic study of when 3D conformer geometry complements 2D molecular fingerprints, yielding three main contributions: (1) an empirical property taxonomy showing that conformer statistics selectively help solvation-dependent properties but not electronic or steric tasks, validated across 14 regression targets with 10-seed significance testing and scaffold splits; (2) a performance hierarchy—SchNet $\approx$ FP+3D > FP+XGBoost > neural conformer ensembles—revealing that the pre-computed feature bottleneck, not the absence of 3D information, limits conformer ensemble methods; and (3) mechanistic understanding of why fingerprints dominate (breadth of chemical information) and when conformer statistics help (solvation properties of large, flexible molecules).

For practitioners, the decision framework is: start with Morgan FP + XGBoost; add conformer $\boldsymbol{\mu}$ statistics for solvation/flexibility properties with >500 molecules; consider physicochemical 3D descriptors (PMI, SASA, USR) for further gains; and use end-to-end 3D GNNs when maximum accuracy justifies the cost. Five to ten conformers suffice for hybrid models.

Limitations. Our conformer features use RDKit ETKDG with uniform (not Boltzmann) weights. The main benchmark uses Murcko scaffold splits; auxiliary experiments (hybrid, learning curves) re-split data independently, so FP-only baselines vary slightly across tables (up to $\sim$6%). The hybrid approach uses simple concatenation; learned feature interaction might yield larger gains. Future work should explore equivariant GNNs beyond SchNet and validate on larger ChEMBL/ZINC datasets.

Reproducibility. All code, model implementations, experiment scripts, and configuration files are publicly available at https://github.com/anonymoussubmitter-167/DKO under the MIT license. The repository includes conformer generation pipelines, all 13 model configurations, hybrid feature extraction, and analysis scripts to reproduce every table and figure in this paper.

References

[1] Pedro J Ballester and W Graham Richards. 2007. Ultrafast Shape Recognition to Search Compound Databases for Similar Molecular Shapes. *Journal of Computational Chemistry* 28, 10 (2007), 1711–1723.

[2] Guy W Bemis and Mark A Murcko. 1996. The Properties of Known Drugs. 1. Molecular Frameworks. *Journal of Medicinal Chemistry* 39, 15 (1996), 2887–2893.

[3] BioRender. 2025. BioRender.com. https://biorender.com Scientific figure created with BioRender.com.

[4] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 785–794.

[5] John S Delaney. 2004. ESOL: Estimating Aqueous Solubility Directly from Molecular Structure. *Journal of Chemical Information and Computer Sciences* 44, 3 (2004), 1000–1005.

[6] Octavian-Eugen Ganea, Lagnajit Pattanaik, Connor W Coley, Regina Barzilay, Klavs F Jensen, William H Green, and Tommi S Jaakkola. 2021. GeoMol: Torsional Geometric Generation of Molecular 3D Conformer Ensembles. In *Advances in Neural Information Processing Systems*, Vol. 34.

[7] Johannes Gasteiger, Shankari Giri, Johannes T Margraf, and Stephan Günnemann. 2020. Fast and Uncertainty-Aware Directional Message Passing for Non-Equilibrium Molecules. In *Machine Learning for Molecules Workshop at NeurIPS*.

[8] Johannes Gasteiger, Janek Groß, and Stephan Günnemann. 2020. Directional Message Passing for Molecular Graphs. In *International Conference on Learning Representations*.

[9] Tobias Gensch, Gabriel dos Passos Gomes, Pascal Friederich, Ellyn Peters, Theresa Gauber, Joseph R Struble, Spencer D Dreher, Franziska Schoenebeck, Matthew S Sigman, and Alán Aspuru-Guzik. 2022. A Comprehensive Discovery Platform for Organophosphorus Ligands for Catalysis. *Journal of the American Chemical Society* 144, 3 (2022), 1205–1217.

[10] Justin Gilmer, Samuel S Schoenholz, Patrick F Riley, Oriol Vinyals, and George E Dahl. 2017. Neural Message Passing for Quantum Chemistry. In *International Conference on Machine Learning*. 1263–1272.

[11] Paul C D Hawkins. 2017. Conformation Generation: The State of the Art. *Journal of Chemical Information and Modeling* 57, 8 (2017), 1747–1756.

[12] Paul C D Hawkins, A Geoffrey Skillman, Gregory L Warren, Benjamin A Ellingson, and Matthew T Stahl. 2010. Conformer Generation with OMEGA: Algorithm and Validation Using High Quality Structures from the Protein Databank and Cambridge Structural Database. *Journal of Chemical Information and Modeling* 50, 4 (2010), 572–584.

[13] Esther Heid, Kevin P Greenman, Yujie Chung, Shih-Cheng Li, David E Graff, Florence H Vermeire, Haoyang Wu, William H Green, and Charles J McGill. 2024. Chemprop: A Machine Learning Package for Chemical Property Prediction. *Journal of Chemical Information and Modeling* 64, 1 (2024), 9–17.

[14] Jarmo Huuskonen. 2000. Estimation of Aqueous Solubility for a Diverse Set of Organic Compounds Based on Molecular Topology. *Journal of Chemical Information and Computer Sciences* 40, 3 (2000), 773–777.

[15] Dejun Jiang, Zhenxing Wu, Chang-Yu Hsieh, Guangyong Chen, Ben Liao, Zhe Wang, Chao Shen, Dongsheng Cao, Jian Wu, and Tingjun Hou. 2021. Could Graph Neural Networks Learn Better Molecular Representation for Drug Discovery? A Comparison Study of Descriptor-based and Graph-based Models. *Journal of Cheminformatics* 13, 1 (2021), 12.

[16] Diederik P Kingma and Jimmy Ba. 2015. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations*.

[17] Alexander Kraskov, Harald Stögbauer, and Peter Grassberger. 2004. Estimating Mutual Information. *Physical Review E* 69, 6 (2004), 066138.

[18] Gregory A Landrum. 2016. RDKit: Open-Source Cheminformatics Software. (2016). https://www.rdkit.org.

[19] Juho Lee, Yoonho Lee, Jungtaek Kim, Adam R Kosiorek, Seungjin Choi, and Yee Whye Teh. 2019. Set Transformer: A Framework for Attention-based Permutation-Invariant Input. In *International Conference on Machine Learning*.

[20] Yi-Lun Liao and Tess Smidt. 2023. Equiformer: Equivariant Graph Attention Transformer for 3D Atomistic Graphs. In *International Conference on Learning Representations*.

[21] Christopher A Lipinski, Franco Lombardo, Beryl W Dominy, Paul J Feeney, et al. 2001. Experimental and Computational Approaches to Estimate Solubility and Permeability in Drug Discovery and Development Settings. *Advanced Drug Delivery Reviews* 46, 1–3 (2001), 3–26.

[22] Ilya Loshchilov and Frank Hutter. 2019. Decoupled Weight Decay Regularization. *International Conference on Learning Representations* (2019).

[23] Simon Mitternacht. 2016. FreeSASA: An Open Source C Library for Solvent Accessible Surface Area Calculations. *F1000Research* 5 (2016), 189.

[24] David L Mobley and J Peter Guthrie. 2014. FreeSolv: A Database of Experimental and Calculated Hydration Free Energies, with Input Files. *Journal of Computer-Aided Molecular Design* 28, 7 (2014), 711–720.

[25] Philipp Pracht, Fabian Bohle, and Stefan Grimme. 2020. Automated Exploration of the Low-Energy Chemical Space with Fast Quantum Chemical Methods. *Physical Chemistry Chemical Physics* 22, 14 (2020), 7169–7192.

[26] Raghunathan Ramakrishnan, Pavlo O Dral, Matthias Rupp, and O Anatole von Lilienfeld. 2014. Quantum Chemistry Structures and Properties of 134 Kilo Molecules. *Scientific Data* 1 (2014), 140022.

[27] Sereina Riniker and Gregory A Landrum. 2015. Better Informed Distance Geometry: Using What We Know To Improve Conformation Generation. *Journal of Chemical Information and Modeling* 55, 12 (2015), 2562–2574.

[28] David Rogers and Mathew Hahn. 2010. Extended-Connectivity Fingerprints. *Journal of Chemical Information and Modeling* 50, 5 (2010), 742–754.

[29] Yu Rong, Yatao Bian, Tingyang Xu, Weiyang Xie, Ying Wei, Wenbing Huang, and Junzhou Huang. 2020. Self-Supervised Graph Transformer on Large-Scale Molecular Data. In *Advances in Neural Information Processing Systems*, Vol. 33.

[30] Matthias Rupp, Alexandre Tkatchenko, Klaus-Robert Müller, and O Anatole von Lilienfeld. 2012. Fast and Accurate Modeling of Molecular Atomization Energies with Machine Learning. *Physical Review Letters* 108, 5 (2012), 058301.

[31] Kristof T Schütt, Pieter-Jan Kindermans, Huziel Enoc Sauceda, Stefan Chmiela, Alexandre Tkatchenko, and Klaus-Robert Müller. 2017. SchNet: A Continuous-Filter Convolutional Neural Network for Modeling Quantum Interactions. In *Advances in Neural Information Processing Systems*, Vol. 30.

[32] Kristof T Schütt, Oliver T Unke, and Michael Gastegger. 2021. Equivariant Message Passing for the Prediction of Tensorial Properties and Molecular Spectra. In *International Conference on Machine Learning*. 9377–9388.

[33] Hannes Stark, Dominique Beaini, Gabriele Corso, Prudencio Tossou, Christian Dallago, Stephan Günnemann, and Pietro Liò. 2022. 3D Infomax Improves GNNs for Molecular Property Prediction. In *International Conference on Machine Learning*.

[34] B L Welch. 1947. The Generalization of 'Student's' Problem when Several Different Population Variances are Involved. *Biometrika* 34, 1/2 (1947), 28–35.

[35] Zhenqin Wu, Bharath Ramsundar, Evan N Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh S Pappu, Karl Leswing, and Vijay Pande. 2018. MoleculeNet: A Benchmark for Molecular Machine Learning. *Chemical Science* 9, 2 (2018), 513–530.

[36] Kevin Yang, Kyle Swanson, Wengong Jin, Connor Coley, Philipp Eiden, Hua Gao, Angel Guzman-Perez, Timothy Hopper, Brian Kelley, Miriam Mathea, et al. 2019. Analyzing Learned Molecular Representations for Property Prediction. *Journal of Chemical Information and Modeling* 59, 8 (2019), 3370–3388.

[37] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan R Salakhutdinov, and Alexander J Smola. 2017. Deep Sets. In *Advances in Neural Information Processing Systems*, Vol. 30.

[38] Gengmo Zhou, Zhifeng Gao, Qiankun Ding, Hang Zheng, Hongteng Xu, Zhewei Wei, Linfeng Zhang, and Guolin Ke. 2023. Uni-Mol: A Universal 3D Molecular Representation Learning Framework. *International Conference on Learning Representations* (2023).

[39] Yanqiao Zhu, Jeehyun Hwang, Keir Adams, Zhen Liu, Bozhao Nan, Brock Stenfors, Yuanqi Du, Jatin Chauhan, Olaf Wiest, Olexandr Isayev, Connor W Coley, Yizhou Sun, and Wei Wang. 2024. Learning Over Molecular Conformer Ensembles: Datasets and Benchmarks. In *International Conference on Learning Representations*.

A Full Per-Dataset Results

Table 10 presents the complete benchmark results for ESOL and FreeSolv with all models, including baseline architectures. Models are described in Section 3.3: **single_conformer** uses only the lowest-energy conformer; **DeepSets** [37] applies permutation-invariant aggregation; **dko_separate_nets** uses independent $\boldsymbol{\mu}/\boldsymbol{\Sigma}$ networks without kernel fusion. Full results for all 6 regression datasets are available in the supplementary materials.

Table 10: Full results on ESOL and FreeSolv (RMSE ± std, 3 seeds).

Model	ESOL	FreeSolv
dko_first_order	1.646±0.036	4.513±0.110
attention	1.888±0.011	4.077±0.071
mean_ensemble	2.016±0.070	4.177±0.048
dko_diagonal	2.040±0.033	4.699±0.057
dko	2.056±0.030	4.881±0.140
dko_separate_nets	2.249±0.031	4.790±0.134
single_conformer	2.394±0.120	4.087±0.100
DeepSets	2.463±0.637	4.277±0.177

B Implementation Sensitivity Analysis

The DKO implementation is sensitive to several hyperparameters, identified through systematic ablation:

(1) **Learning rate** (10^{-5} vs 10^{-4}): Using a 10× lower learning rate than other models reduced ESOL RMSE by 30% (3.318 → 2.309) when corrected.
(2) **Feature normalization** (dim=$(1, 2)$ vs dim=1): Normalizing across both conformers and features destroyed inter-feature variance in $\boldsymbol{\Sigma}$, making all covariance features approximately identical. Correcting to dim=1 restored Pearson correlation from ∼0 to ∼0.48.
(3) **Kernel output dimension** (32 vs 64): Marginal individual effect but needed for combined benefit.
(4) **Sigma regularization** (10^{-4} vs 10^{-2}): Too-small regularization caused near-singular covariance matrices.
(5) **Mixed precision**: FP16 arithmetic is insufficient for covariance matrix operations. Enabling mixed precision increased RMSE by 30% with 10× higher variance.

Combined, these corrections improved DKO's ESOL R^2 from −1.70 to −0.04, a swing of 1.66 in explained variance. The sensitivity of covariance-based methods to numerical precision and normalization choices is an important practical consideration.

C Conformer Diversity Analysis

Table 11 reports conformer diversity metric (CDM) statistics per dataset. CDM values are specific to our feature encoding and not directly transferable to other pipelines.

Table 11: Conformer diversity metric (CDM = $\mathrm{tr}(\boldsymbol{\Sigma})$) statistics by dataset. Values are feature-encoding-specific.

Dataset	Median	Max	Q1	Q3
FreeSolv	0.01	42.8	0.0	24.95
ESOL	27.88	148.04	0.0	67.32
QM9	25.05	113.76	13.73	49.08
Lipophilicity	69.99	155.53	34.80	98.83

Lipophilicity has 3× higher median CDM than QM9, consistent with dko_invariants achieving its strongest results on this dataset. FreeSolv has near-zero median CDM yet still benefits from hybrid features—primarily through first-order ($\boldsymbol{\mu}$) statistics (Table 5), confirming that CDM does not predict first-order conformer utility.

D Enhanced 3D Descriptors Benchmark

Table 12 presents the full benchmark of enhanced 3D descriptors across all feature combinations and datasets (mean ± std, 3 seeds). The 28 descriptors span four categories: **Shape** (9): 2 PMI ratios, 3 raw principal moments, asphericity, eccentricity, radius of gyration, inertial shape factor; **Surface** (1): total SASA; **USR** (12): distances from centroid, closest/farthest atom references; **Global** (6): span, molecular volume, compactness, 3 bounding box extents.

Table 12: Full 3D descriptor benchmark (RMSE, mean of 3 seeds). "3D" = 28 physicochemical descriptors; "Geo" = geometric $\boldsymbol{\mu}$+$\boldsymbol{\Sigma}$ features.

Features	ESOL	FreeSolv	Lipo	QM9-Gap
3D-only (μ)	1.340	4.053	1.119	0.035
3D μ+σ	1.359	4.100	1.094	0.034
FP only	1.507	2.939	0.910	0.020
FP + 3D μ	1.025	2.936	0.868	0.020
FP + 3D μ+σ	1.000	3.073	0.873	0.020
FP + Geo μ+σ	1.358	2.824	0.957	0.021
FP + 3D + Geo	1.094	2.597	0.916	0.020

Key observations: (i) 3D descriptors alone ($R^2 \approx 0.55$ on ESOL) outperform raw geometric $\boldsymbol{\mu}$ features, validating the physicochemical relevance of PMI/SASA/USR; (ii) combining FP+3D achieves the best non-GNN result on ESOL (1.000, $R^2 = 0.75$); (iii) adding geometric features to FP+3D helps on FreeSolv (2.597 vs. 3.073) but not ESOL (1.094 vs. 1.000), suggesting dataset-specific feature complementarity.

E Conformer Count Ablation

Table 13 shows the effect of conformer count on ESOL for both XGBoost (hybrid) and dko_gated (neural) models.

Table 13: Conformer count ablation on ESOL (mean of 3 seeds). XGBoost saturates at n=5–10; dko_gated improves monotonically.

n	XGB RMSE	XGB R^2	DKO RMSE	DKO R^2
1	1.401	0.519	5.487	<0
5	1.337	0.562	1.773	0.229
10	1.330	0.567	1.723	0.272
20	1.354	0.550	1.704	0.288
50	1.358	0.548	1.642	0.339

XGBoost reaches near-optimal performance at n=5–10 conformers (RMSE 1.330–1.337) with slight degradation at higher counts, likely due to noise in conformer mean statistics. In contrast, dko_gated improves monotonically from degenerate (n=1, $R^2 < 0$) to $R^2 = 0.34$ at n=50, suggesting the neural architecture benefits from richer distributional estimates. Practical implication: for XGBoost-based hybrid models, generating 5–10 conformers is sufficient, saving significant computational cost vs. the default n=50.

F SchNet Architecture Details

We use SchNet [31] with 128 hidden channels, 6 interaction blocks, 50 radial basis function (RBF) Gaussians, and a 10Å cutoff radius. Each molecule is represented by up to 10 conformers (lowest-energy from ETKDG); predictions are averaged across conformers. Training uses Adam [16] (not AdamW, as SchNet's original implementation uses Adam) with learning rate 5×10^{-4}, batch size 32, and early stopping (patience 50). Per-seed results are shown in Table 14.

Table 14: SchNet per-seed results (RMSE / R^2, 3 seeds).

Seed	ESOL	FreeSolv	Lipo
42	1.017 / 0.747	2.591 / 0.517	0.721 / 0.611
123	0.943 / 0.782	2.055 / 0.696	0.714 / 0.618
456	1.054 / 0.728	2.327 / 0.611	0.713 / 0.620
Mean	1.004 / 0.752	2.324 / 0.608	0.716 / 0.616

FreeSolv shows the highest variance across seeds (std = 0.268), consistent with its small dataset size (642 molecules). SchNet's strong performance despite averaging over only 10 conformers (not ensemble statistics) suggests that end-to-end 3D learning captures geometric information more effectively than pre-computed summary statistics.

G Hybrid Neural MLP vs XGBoost

We compare a neural MLP (2309→512→256→128→1, ReLU activations, dropout 0.1) against XGBoost on the same hybrid features (FP + $\boldsymbol{\mu}$ + $\boldsymbol{\Sigma}$).

Table 15: Hybrid MLP vs. XGBoost (RMSE / R^2, mean of 3 seeds).

Method	ESOL	FreeSolv	Lipo	QM9-Gap
MLP	1.218 / .637	3.809 / −.044	0.906 / .394	.023 / .787
XGBoost	1.358 / .548	2.824 / .423	0.957 / .325	.021 / .821

MLP outperforms XGBoost on ESOL (1.218 vs. 1.358) and Lipophilicity (0.906 vs. 0.957), but XGBoost wins on FreeSolv (2.824 vs. 3.809) and QM9-Gap (0.021 vs. 0.023). The MLP's negative R^2 on FreeSolv indicates overfitting on this small dataset (642 molecules), while XGBoost's built-in regularization prevents this failure mode. This suggests that the optimal learning algorithm for hybrid features is dataset-dependent.

H Mutual Information Analysis

We estimate mutual information (MI) using the Kraskov–Stögbauer–Grassberger (KSG) k-nearest-neighbor estimator [17] with k=5, selecting the top 50 features by individual MI for each feature group. For MI estimation, we project the 1024-dimensional $\boldsymbol{\mu}$ to 256 dimensions via PCA to mitigate the curse of dimensionality in KSG k-nearest-neighbor estimation.

Table 16: Mutual information analysis (nats, top-50 features per group). "Per-feat." = mean MI per feature.

Dataset	Group	Total MI	Per-feat.
ESOL	FP (2048-bit)	1.10	0.022
	$\boldsymbol{\mu}$ (256-dim PCA)	8.27	0.165
	$\boldsymbol{\Sigma}$ (5-dim)	0.35	0.070
FreeSolv	FP	2.28	0.046
	$\boldsymbol{\mu}$	4.39	0.088
	$\boldsymbol{\Sigma}$	0.22	0.045
Lipo	FP	0.38	0.008
	$\boldsymbol{\mu}$	0.75	0.015
	$\boldsymbol{\Sigma}$	0.03	0.006
QM9-Gap	FP	1.79	0.036
	$\boldsymbol{\mu}$	9.70	0.194
	$\boldsymbol{\Sigma}$	0.54	0.108

Key findings: (i) $\boldsymbol{\mu}$ features carry 2–8× more MI per feature than FP bits across all datasets; (ii) despite higher per-feature informativeness, $\boldsymbol{\mu}$'s advantage depends on dataset—ESOL shows 7.5× ratio while FreeSolv shows only 1.9×; (iii) $\boldsymbol{\Sigma}$ features have moderate per-feature MI (0.006–0.108 nats) but contribute minimal *conditional* MI given $\boldsymbol{\mu}$ (−1.5% to −3.5%, Table 17).

Table 17: Conditional MI of $\boldsymbol{\Sigma}$ given $\boldsymbol{\mu}$ (nats). Negative values indicate $\boldsymbol{\Sigma}$ is redundant given $\boldsymbol{\mu}$.

Dataset	MI(μ)	MI(μ+Σ)	Δ%
ESOL	8.27	7.98	−3.5%
FreeSolv	4.39	4.32	−1.5%
Lipo	0.75	0.74	−2.2%
QM9-Gap	9.70	9.39	−3.2%

Limitations: KSG MI estimation is sensitive to k and dimensionality. The variance-based top-50 feature selection favors continuous features ($\boldsymbol{\mu}$) over binary FP bits, so joint MI estimates for combined feature sets effectively measure $\boldsymbol{\mu}$ MI only; the conditional MI in Table 17 therefore conditions on $\boldsymbol{\mu}$ alone rather than FP+$\boldsymbol{\mu}$. Future work should use MI-ranked or per-group proportional feature selection to properly estimate joint MI across heterogeneous feature types.

I Feature Attribution

We report XGBoost feature importance (gain-based) aggregated by feature group across 3 seeds.

Top-10 features by dataset. On ESOL, 7 of the top 10 features are $\boldsymbol{\mu}$ dimensions (mu_dim_137, mu_dim_138,

Table 18: Feature group importance (% of total gain). $\boldsymbol{\Sigma}$ contributes <2% across all datasets.

Dataset	FP (%)	$\boldsymbol{\mu}$ (%)	$\boldsymbol{\Sigma}$ (%)
ESOL	31.4	67.4	1.2
FreeSolv	62.0	36.7	1.3
QM9-Gap	49.3	49.0	1.7

mu_dim_131 are top 3), reflecting that mean conformer geometry is the primary signal for solubility. On FreeSolv, all top 10 are fingerprint bits (FP_bit_314, FP_bit_807, FP_bit_1114), with $\boldsymbol{\mu}$ features first appearing at rank 15. On QM9-Gap, the top 10 alternate between $\boldsymbol{\mu}$ (rank 1: mu_dim_146) and FP (rank 2: FP_bit_1380), reflecting the balanced importance.

Among the 5 $\boldsymbol{\Sigma}$ features, top5_var and effective_rank contribute the most across datasets, while total_var and max_var are generally less important. The sigma feature sigma_mean_var appears at rank 17 on QM9-Gap (importance 0.85%), the highest-ranked $\boldsymbol{\Sigma}$ feature on any dataset. Note that the hybrid XGBoost model uses variance-based $\boldsymbol{\Sigma}$ features ($\boldsymbol{\sigma}_5$; Section 3.5), while dko_invariants uses matrix-invariant features (Section 3.3).

J Scaffold Split Validation Details

Table 19 presents the full scaffold split results including Mu-only features.

Table 19: Full scaffold vs. random split results (RMSE, mean ± std, 3 seeds).

Dataset	Split	FP-only	FP+μ+Σ	μ-only
ESOL	random	1.083±0.057	0.991±0.083	1.315±0.056
	scaffold	1.492±0.016	1.315±0.019	1.595±0.002
FreeSolv	random	1.701±0.093	2.008±0.199	3.011±0.261
	scaffold	3.107±0.114	3.173±0.154	3.935±0.147
Lipo	random	0.881±0.028	0.887±0.039	1.099±0.033
	scaffold	0.914±0.001	0.937±0.006	1.097±0.012

Scaffold splits are substantially harder than random splits across all datasets and feature configurations, with FP-only RMSE increasing by 38% (ESOL), 83% (FreeSolv), and 4% (Lipophilicity). The small Lipophilicity degradation suggests that octanol–water partition coefficients generalize well across scaffolds, consistent with this property being primarily determined by local functional group contributions. Mu-only features consistently underperform FP-only under both split types, confirming that conformer features alone cannot compete with fingerprints for generalization.

K Learning Curve Details

Table 20 presents the full learning curve results with absolute RMSE values.

The opposite scaling patterns across datasets are striking: on ESOL and FreeSolv, hybrid features become increasingly valuable with more data (suggesting genuine signal), while on Lipophilicity and QM9-Gap, the penalty diminishes (suggesting XGBoost learns to downweight irrelevant features). The FreeSolv crossover at ∼75% (∼440 molecules) provides a practical threshold for when conformer features become beneficial on small solvation datasets.

Table 20: Learning curves: absolute RMSE (mean of 3 seeds) at each training fraction.

	ESOL		FreeSolv		Lipo		QM9-Gap	
Frac	FP	Hyb	FP	Hyb	FP	Hyb	FP	Hyb
10%	2.007	1.913	3.597	4.096	1.066	1.131	.0241	.0279
25%	1.744	1.665	3.135	3.592	0.966	1.056	.0219	.0244
50%	1.597	1.521	3.221	3.328	0.950	0.985	.0209	.0224
75%	1.578	1.419	3.056	2.988	0.928	0.956	.0205	.0215
100%	1.533	1.343	3.077	2.862	0.920	0.932	.0201	.0208

L Error Analysis Details

FreeSolv (overall: −2.8%): Larger molecules benefit from hybrid features (Q4 heavy atoms: +7.0%, Q4 molecular weight: +7.4%) while small molecules are hurt (Q1 molecular weight: −34.5%). Molecular weight significantly correlates with hybrid improvement (Spearman $r = 0.30$, $p = 0.025$), as does heavy atom count ($r = 0.28$, $p = 0.042$).

Lipophilicity (overall: −5.7%): Hybrid features hurt uniformly across all quartiles of all molecular properties. No significant correlations between any molecular descriptor and hybrid improvement (all $p > 0.29$). This confirms that conformer features are genuinely uninformative for lipophilicity prediction, regardless of molecular size, flexibility, or surface properties.

ESOL TPSA analysis: Improvement is relatively uniform across TPSA quartiles (+3.9% to +13.6%), suggesting that hybrid benefit is not driven by polar surface area. The strongest signals are molecular size (heavy atoms, MW) and flexibility (rotatable bonds), both of which directly determine conformer diversity.

M MARCEL Benchmark Details

Table 21: Kraken steric descriptors (RMSE, mean of 3 seeds, random 80/10/10 splits; std omitted for clarity, all <6% of mean). Bold: best overall per column. *Italic*: best neural method per column.

Model	B5	L	burB5	burL
FP+XGB	**0.519**	**0.650**	**0.378**	**0.259**
attention	1.030	1.233	*0.559*	*0.356*
mean	*0.982*	*1.228*	0.592	0.387
dko_gated	1.176	1.322	0.653	0.409

On Kraken steric descriptors (Table 21), which are Boltzmann-averaged 3D properties, all neural conformer methods lag behind FP+XGBoost. Among neural methods, attention and mean aggregation achieve comparable performance: attention wins on buried descriptors (burB5, burL) while mean aggregation wins on standard descriptors (B5, L). DKO's distributional summary performs worst on all 4 targets, suggesting that covariance statistics are not well-suited for Boltzmann-averaged steric properties where per-conformer detail matters.

On BDE, FP+XGBoost achieves RMSE 4.833 ($R^2 = 0.958$) while all neural conformer models essentially predict the mean ($R^2 \approx 0$), a ∼5× RMSE gap. Three factors explain the failure: (i) mean Pearson feature–target correlation is only $|\bar{r}| = 0.044$ (near-random); (ii) variable feature dimensions across molecules (187–1854) require padding/truncation; (iii) the conformer generation pipeline uses uniform rather than MMFF94-derived Boltzmann weights. On Drugs-75K electronic properties, FP+XGBoost wins by 21–29%, further supporting this property-type dependence.

N 10-Seed Neural Model Comparison

Table 22: 10-seed statistical validation (ESOL and Lipophilicity). ± denotes standard deviation across seeds. One-sided Welch's t-test. "n.s.": not significant ($p > 0.05$); "—": reference model.

Dataset	Model	RMSE	Δ
ESOL	dko_gated	1.654±0.032	−12.1%
	dko_inv.	1.765±0.050	−6.2%
	attention	1.881±0.027	—
Lipophilicity	dko_inv.	1.140±0.008	n.s.
	attention	1.140±0.006	—
	dko_gated	1.164±0.022	n.s.

Welch's t-test [34] confirms that dko_gated significantly outperforms attention on ESOL ($p < 0.001$, 12.1% improvement), while Lipophilicity differences are not significant ($p > 0.05$).

O ESOL Error Stratification

Table 23: ESOL hybrid improvement by molecular property quartile (single seed). Q1 = smallest/fewest, Q4 = largest/most.

Property	Q1	Q2	Q3	Q4
Heavy atoms	+10.8%	+3.8%	+0.1%	**+18.9%**
Rotatable bonds	−0.3%	+5.0%	**+39.7%**	+11.0%
Mol. weight	+8.0%	−2.0%	−0.3%	**+23.9%**
TPSA	+3.9%	+11.0%	+13.6%	+13.4%

The largest improvements occur for molecules with >22 heavy atoms (Q4: +18.9%), 1–2 rotatable bonds (Q3: +39.7%), and molecular weight >300 Da (Q4: +23.9%). Rigid molecules with zero rotatable bonds (Q1, n=49) show essentially no benefit (−0.3%). On FreeSolv, molecular weight significantly correlates with hybrid improvement (Spearman $r = 0.30$, $p = 0.025$).

P Learning Curve Details (Percentage Improvement)

Table 24: Learning curves: hybrid improvement (%) at different training data fractions. Positive values indicate hybrid is better.

Fraction	ESOL	FreeSolv	Lipo	QM9-Gap
10%	+4.7%	−13.9%	−6.2%	−15.8%
25%	+4.5%	−14.6%	−9.3%	−11.5%
50%	+4.8%	−3.3%	−3.7%	−7.4%
75%	+10.1%	+2.2%	−3.0%	−5.2%
100%	+12.4%	+7.0%	−1.3%	−3.4%

On ESOL, hybrid improvement grows monotonically from +4.7% at 10% data to +12.4% at 100%, suggesting conformer features add genuine signal. On FreeSolv, the pattern is dramatic: hybrid features hurt at small data sizes (−13.9% at 10%) but become beneficial with sufficient data (+7.0% at 100%), with a crossover at ∼75% (∼440 molecules). On Lipophilicity and QM9-Gap, conformer features consistently hurt but the penalty diminishes with more data.

Q Ablation Details

Eigenvalue concentration. The feature variance audit reveals that the top-10 eigenvalues of $\mathbf{\Sigma}$ capture only 4–8% of total variance across datasets. For comparison, uniform variance across D=1024 dimensions would yield <1% in the top 10; the moderate concentration suggests structure exists but is too distributed for low-rank methods. The effective rank at 90% explained variance ranges from 353 (QM9) to 685 (Lipophilicity), explaining why lowrank (average rank 9.25) and eigenspectrum (rank 7.25) variants underperform invariants (rank 4.50): richer representations overfit on datasets of 642–4,200 molecules.

Synthetic validation. We generate N=2,000 synthetic molecules with D=128 features, n=50 conformers, and ground-truth labels $y = \mathbf{W}^\top \boldsymbol{\mu} + \alpha \log(1 + \mathrm{tr}(\mathbf{\Sigma}))$, varying $\alpha \in \{0, 0.1, 0.5, 1.0\}$. Even at α=0 (no covariance signal), the kernel DKO achieves 27–30% lower RMSE than first-order baselines, demonstrating a beneficial inductive bias from the kernel parameterization itself.